29/01/2025
Evening

Aakash
Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1

[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Let a circle C pass through the points (4, 2) and (0, 2), and its centre lie on $3x + 2y + 2 = 0$ Then the length of the chord, of the circle C, whose mid-point is (1, 2), is:

(1) $2\sqrt{3}$

(2) $\sqrt{3}$

(3) $4\sqrt{2}$

(4) $2\sqrt{2}$

Answer (1)

Sol. Let the centre be

$$\left(-2a, \frac{6a-2}{2}\right) \equiv (-2a, 3a-1)$$

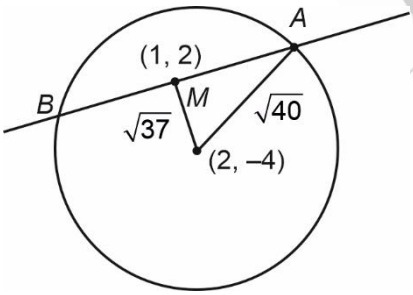
Centre is equal distance from (4, 2) and (0, 2)

$$\Rightarrow \sqrt{(4+2a)^2+(3a-3)^2} = \sqrt{(-2a-0)^2+(3a-3)^2}$$

$$\Rightarrow (2a+4)^2 + 9(a-1)^2 = 4a^2 + 9(a-1)^2$$

$$\Rightarrow 4a^2 + 16 + 16a = 4a^2 \Rightarrow a = -1$$

$$\Rightarrow \text{centre} \equiv (2,-4) \Rightarrow \text{Radius} = \sqrt{40}$$



$$\Rightarrow AM^2 = \left(\sqrt{40}\right)^2 - \left(\sqrt{37}\right)^2$$

$$\Rightarrow 2AM = AB = 2\sqrt{3}$$

2. Let $f(x) = \int_0^x t(t^2 - 9t + 20)dt$, $1 \le x \le 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha + \beta)$ equals

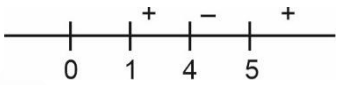
(1) 125 (2) 253

(3) 154 (4) 157

Answer (4)

Sol. $f'(x) = x(x^2 - 9x + 20)$, $x \in (1, 5)$

$= (x - 4)x(x - 5)$



$\Rightarrow f'(x) > 0 \forall x \in (1, 4)$

$\Rightarrow f'(x) < 0 \forall x \in (4, 5)$

$\Rightarrow$ $f(x)$ increasing in (1, 4)

$f(x)$ decreasing in (4, 5)

$\Rightarrow$ critical points to check:

$x = 1, 4, 5$

$$f(x) = \int_0^x (t^3 - 9t^2 + 20t)dt$$

$$= \frac{t^4}{4} - 3t^3 + 10t^2\Big|_0^x = \frac{x^4}{4} - 3x^3 + 10x^2$$

$$f(1) = \frac{1}{4} - 3 + 10 = \frac{29}{4}$$

$$f(4) = 4^3 - 3.4^3 + 10.4^2 = -2.4^3 + 10.4^2 = 32$$

$$f(5) = \frac{5^4}{4} - 3.5^3 + 10.25 = \frac{5^4}{4} - 125 = \frac{125}{4}$$

Range $\Rightarrow \left[\frac{29}{4}, 32\right] \Rightarrow 4(\alpha+\beta) = 128 + 29$

$= 157$



3. Let the function $f(x)=(x^2-1)\left|x^2-ax+2\right|+\cos|x|$ be not differentiable at the two points $x=\alpha=2$ and $x=\beta$. Then the distance of the point (α, β) from the line $12x+5y+10=0$ is equal to

(1) 3 (2) 2

(3) 5 (4) 4

Answer (1 *Bonus)

Sol. $f(x)=(x^2-1)|x^2-ax+2|+\cos|x|$

Notice that $\cos(-x)=\cos x=\cos|x|$ which means cos|x| is differentiable

everywhere in $x \in R$

$\Rightarrow$ $f(x)$ can be non differentiable where $|x^2-ax+2|=0$

$\Rightarrow$ $x^2-ax+2=0$ $\begin{cases} x=\alpha=2 \\ x=\beta \end{cases}$

$\Rightarrow$ $4-2a+2=0 \Rightarrow a=3$

$\Rightarrow$ $(x^2-3x+2)=0$ $\Rightarrow$ $x=1, 2$

$\beta=1$

but $f(x)$ is differentiable at $x=1$.

so it should be bonus.

distance of (α, β) from line

$12x+5y+10=0$

$\Rightarrow \dfrac{|2(12)+5(1)+10|}{13}=\dfrac{39}{13}=3$

4. Let $A=\left[a_{ij}\right]$ be a matrix of order 3 × 3, with $a_{ij}=\left(\sqrt{2}\right)^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha+\beta\sqrt{2}, \alpha, \beta \in Z$, then $\alpha+\beta$ is equal to

(1) 280 (2) 210

(3) 224 (4) 168

Answer (3)

Sol. $a_{ij}=\left(\sqrt{2}\right)^{i+j}$

$$A^2=\begin{bmatrix} 2 & 2\sqrt{2} & 4 \\ 2\sqrt{2} & 4 & 4\sqrt{2} \\ 4 & 4\sqrt{2} & 8 \end{bmatrix}\times\begin{bmatrix} 2 & 2\sqrt{2} & 4 \\ 2\sqrt{2} & 4 & 4\sqrt{2} \\ 4 & 4\sqrt{2} & 8 \end{bmatrix}$$

$$A^2=\begin{bmatrix} 28 & 28\sqrt{2} & 56 \\ 28\sqrt{2} & 56 & 56\sqrt{2} \\ 56 & 56\sqrt{2} & 112 \end{bmatrix}$$

Sum of elements of third row = $56+112+56\sqrt{2}$

$=168+56\sqrt{2}$

$\Rightarrow \alpha=168$ $\beta=56$

$\therefore$ $\alpha+\beta=224$

5. If for the solution curve $y=f(x)$ of the differential equation $\dfrac{dy}{dx}+(\tan x)y=\dfrac{2+\sec x}{(1+2\sec x)^2}$, $x \in \left(\dfrac{-\pi}{2}, \dfrac{\pi}{2}\right)$, $f\left(\dfrac{\pi}{3}\right)=\dfrac{\sqrt{3}}{10}$, then $f\left(\dfrac{\pi}{4}\right)$ is equal to

(1) $\dfrac{\sqrt{3}+1}{10(4+\sqrt{3})}$ (2) $\dfrac{4-\sqrt{2}}{14}$

(3) $\dfrac{9\sqrt{3}+3}{10(4+\sqrt{3})}$ (4) $\dfrac{5-\sqrt{3}}{2\sqrt{2}}$

Answer (2)

Sol. $\dfrac{dy}{dx}+(\tan x)y=\dfrac{2+\sec x}{(1+2\sec x)^2}$

$IF=e^{\int \tan x\,dx}=e^{\ln \sec x}$

$=\sec x$

$\therefore$ solution will be

$$y\sec x=\int \frac{(2+\sec x)}{(1+2\sec x)^2}\sec x\,dx$$

$$=\int \frac{\left(2+\dfrac{1+\tan^2\frac{x}{2}}{1-\tan^2\frac{x}{2}}\right)\left(1+\tan^2\frac{x}{2}\right)}{\left(1+\dfrac{2\left(1+\tan^2\frac{x}{2}\right)}{1-\tan^2\frac{x}{2}}\right)^2}\left(1-\tan^2\frac{x}{2}\right)dx$$



Let $\tan\frac{x}{2}=t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx=dt$

$$=2\int\frac{\left(3-t^2\right)}{\left(3+t^2\right)^2}dt$$

$$=\frac{2t}{t^2+3}=\frac{2\tan\frac{x}{2}}{\tan^2\frac{x}{2}+3}+c$$

$$\Rightarrow y\sec x=\frac{2\tan\frac{x}{2}}{\tan^2\frac{x}{2}+3}+c$$

$$y\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}}{10}$$

$$\Rightarrow 2y=\frac{\frac{2}{\sqrt{3}}}{\frac{1}{3}+3}+c=\frac{\frac{2}{\sqrt{3}}}{\frac{10}{3}}+c=\frac{2\sqrt{3}}{10}+c$$

$$\frac{2\sqrt{3}}{10}=\frac{2\sqrt{3}}{10}+c\Rightarrow c=0$$

$$\therefore\quad y=\frac{2\tan\frac{x}{2}}{\sec x\left[\tan^2\frac{x}{2}+3\right]}$$

Now, $f\left(\frac{\pi}{4}\right)=\frac{2\tan\frac{\pi}{8}}{\sec\frac{\pi}{4}\left[\tan^2\frac{\pi}{8}+3\right]}=\frac{4-2\sqrt{2}}{14}$

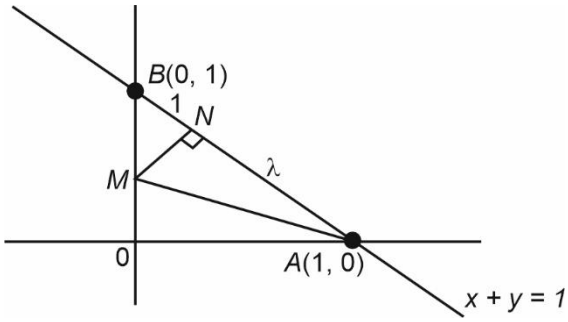
6. Let the line $x + y = 1$ meet the axes of x and y at A and B, respectively. A right angled triangle AMN is inscribed in the triangle OAB, where O is the origin and the points M and N lie on the lines OB and AB, respectively. If the area of the triangle AMN is $\frac{4}{9}$ of the area of the triangle OAB and $AN : NB = \lambda : 1$, then the sum of all possible value(s) of is λ :

(1) $\frac{13}{6}$ (2) $\frac{5}{2}$

(3) $\frac{1}{2}$ (4) 2

Answer (4)

Sol.



$$\frac{AN}{NB}=\frac{\lambda}{1}$$

Then $N\equiv\left(\frac{1}{1+\lambda},\frac{\lambda}{1+\lambda}\right)$

$m_{MN} = 1$

$$MN:\left(y-\frac{\lambda}{1+\lambda}\right)=\left(x-\frac{1}{1+\lambda}\right)$$

$$\therefore\ M\left(0,\frac{\lambda-1}{\lambda+1}\right)$$

area $(\Delta AMN) = \frac{4}{9}$ ar (ΔOAB)

$$\frac{1}{2}\times|AN\times NM|=\frac{4}{9}\times\frac{1}{2}\times1\times1$$

$$|AN\times NM|=\frac{4}{9}$$

$$\left|\frac{\sqrt{2}\lambda}{1+\lambda}\times\frac{\sqrt{2}}{1+\lambda}\right|=\frac{4}{9}$$

$$\left|\frac{2\lambda}{(1+\lambda)^2}\right|=\frac{4}{9}$$

$9|\lambda| = 2(1 + \lambda)^2$

$\pm 9\lambda = 2 + 2\lambda^2 + 4\lambda$

$\Rightarrow 2\lambda^2 + 13\lambda + 2 = 0 \quad \alpha, \beta$



$\Rightarrow 2\lambda^2 - 5\lambda + 2 = 0 \Rightarrow \lambda = \frac{1}{2}, 2$

$\because \quad M\left(0, \frac{\lambda-1}{\lambda+1}\right)$

$\because$ M lies between (0, 0) and (0, 1)

$\Rightarrow \quad \lambda \neq \frac{1}{2}, \alpha, \beta$

$\therefore \quad \lambda = 2$

$\therefore$ Only possible value of $\lambda = 2$

7. If the domain of the function $\log_5(18x - x^2 - 77)$ is (α, β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ, δ), then $\alpha^2+\beta^2+\gamma^2$ is equal to

(1) 174 (2) 195

(3) 179 (4) 186

Answer (4)

Sol. $f_1(x) = \log_5(18x - x^2 - 77)$

$\therefore \quad 18x - x^2 - 77 > 0$

$x^2 - 18x + 77 < 0$

$x \in (7, 11)$

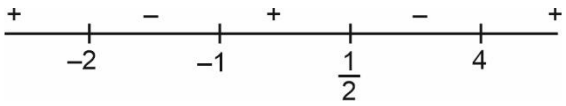
$\alpha = 7, \beta = 11$

$f_2(x) = \log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$

$x > 1, x - 1 \neq 1, \frac{2x^2+3x-2}{x^2-3x-4} > 0$

$x > 1, \ x \neq 2, \frac{(2x-1)(x+2)}{(x-4)(x+1)} > 0$

$x > 1, x \neq 2,$



$\therefore \quad x \in (4, \infty)$

$\therefore \quad \gamma = 4$

$\therefore \quad \alpha^2 + \beta^2 + \gamma^2 = 49 + 121 + 16$

$= 186$

8. Let a straight line L pass through the point $P(2, -1, 3)$ and be perpendicular to the lines $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z-3}{-2}$ and $\frac{x-3}{1} = \frac{y-2}{3} = \frac{z+2}{4}$. If the line L intersects the yz-plane at the point Q, then the distance between the points P and Q is

(1) $\sqrt{10}$ (2) 3

(3) 2 (4) $2\sqrt{3}$

Answer (2)

Sol. Vector parallel to L

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 3 & 4 \end{vmatrix} = 10\hat{i} - 10\hat{j} + 5\hat{k}$$

$= 5\left(2\hat{i} - 2\hat{j} + \hat{k}\right)$

Equation of 'L'

$\frac{x-2}{2} = \frac{y+1}{-2} = \frac{z-3}{1} = \lambda(\text{say})$

Let $Q(2\lambda+2, -2\lambda-1, \lambda+3)$

$\Rightarrow \quad 2\lambda + 2 = 0 \Rightarrow \lambda = -1$

$\Rightarrow \quad Q(0, 1, 2)$

$d(P, Q) = 3$

9. If all the words with or without meaning made using all the letters of the word "KANPUR" are arranged as in a dictionary, then the word at 440th position in this arrangement, is:

(1) PRNAKU (2) PRNAUK

(3) PRKANU (4) PRKAUN

Answer (4)

Sol. A, K, N, P, R, U

A ..... 5! = 120

K ..... 5! = 120

N ..... 5! = 120

P A ..... 4! = 24

P K ..... 4! = 24



P N ..... 4! = 24

P R A .....3! = 6

P R K A N U = 1

P R K A U N = 1

Total = 440

$\Rightarrow$ 440th word is P R K A U N

10. Let $S = N \cup \{0\}$. Define a relation R from S to **R** by:

$$R = \left\{(x, y) : \log_e y = x\log_e\left(\frac{2}{5}\right), x \in S, y \in \mathbf{R}\right\}.$$

Then, the sum of all the elements in the range of R is equal to

(1) $\frac{10}{9}$ (2) $\frac{5}{3}$

(3) $\frac{3}{2}$ (4) $\frac{5}{2}$

Answer (2)

Sol. $\log_e y = x\log_e\left(\frac{2}{5}\right)$

$y = \left(\frac{2}{5}\right)^x$

$x \in N \cup \{0\}$

$y = 1, \frac{2}{5}, \left(\frac{2}{5}\right)^2, \ldots$

$\sum y = \frac{1}{1-\frac{2}{5}} = \frac{5}{3}$

11. Bag 1 contains 4 white balls and 5 black balls, and Bag 2 contains n white balls and 3 black balls. One ball is drawn randomly from Bag 1 and transferred to Bag 2. A ball is then drawn randomly from Bag 2. If the probability, that the ball drawn is white, is 29/45, then n is equal to

(1) 6 (2) 3

(3) 5 (4) 4

Answer (1)

Sol. Bag 1 $\rightarrow 4w, 5B$

Bag 2 $\rightarrow nw, 3B$

(I) $\rightarrow$ Transferred ball is white $P(w) = \frac{n+1}{n+4} \cdot \frac{4}{9}$

(II) $\rightarrow$ Transferred ball is black $P(w) = \frac{5}{9} \cdot \frac{n}{n+4}$

$$\frac{4n+4}{9n+36} + \frac{5n}{9n+36} = \frac{29}{45}$$

$$\frac{9n+4}{9n+36} = \frac{29}{45} \Rightarrow n = 6$$

12. If $\alpha x + \beta y = 109$ is the equation of the chord of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, whose mid point is $\left(\frac{5}{2}, \frac{1}{2}\right)$, then $\alpha + \beta$ is equal to

(1) 37 (2) 58

(3) 72 (4) 46

Answer (2)

Sol. Chord with given mid-point

$T = S_1$

$$\frac{5}{18}x + \frac{y}{8} = \frac{25}{36} + \frac{1}{16} = \frac{109}{144}$$

$40x + 18y = 109$

$\alpha + \beta = 40 + 18 = 58$

13. Let $A = [a_{ij}]$ be a 2 × 2 matrix such that $a_{ij} \in \{0, 1\}$ for all i and j. Let the random variable X denote the possible values of the determinant of the matrix A. Then, the variance of X is:

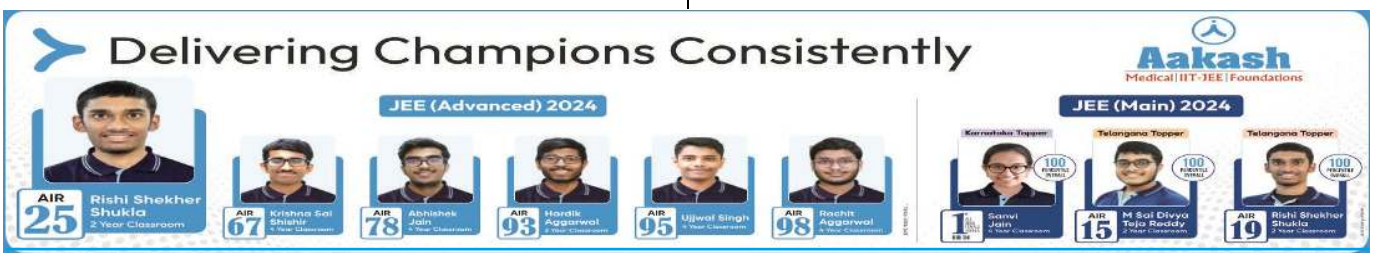
(1) $\frac{3}{8}$ (2) $\frac{5}{8}$

(3) $\frac{1}{4}$ (4) $\frac{3}{4}$

Answer (1)

Sol.

x	0	1	–1
$P(x)$	$\frac{10}{16}$	$\frac{3}{16}$	$\frac{3}{16}$



$\text{Var}(x) = E(x^2) - [E(x)]^2$

$= \sum_{i=1}^{3} x_i^2 P(x_i) - (\mu)^2$

$= 1\times\frac{3}{16} + 1\times\frac{3}{16} \quad [\mu = 0]$

$= \frac{6}{16} = \frac{3}{8}$

14. Let P be the foot of the perpendicular from the point $(1, 2, 2)$ on the line $L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$. Let the $\vec{r} = \left(-\hat{i} + \hat{j} - 2\hat{k}\right) + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right),\ \lambda \in \mathbf{R}$, intersect the line L at Q. Then $2(PQ)^2$ is

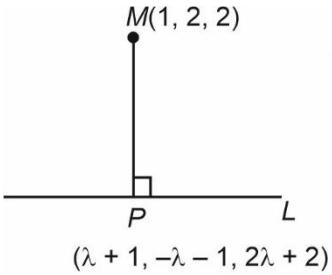
(1) 29 (2) 27

(3) 25 (4) 19

Answer (2)

Sol. General point on line L : $\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$

is $(\lambda + 1, -\lambda - 1, 2\lambda + 2)$



DR's of PM are $(\lambda, -\lambda - 3, 2\lambda)$

$PM \perp L$

$\Rightarrow \lambda + (-1)(-\lambda - 3) + 2(2\lambda) = 0$

$\Rightarrow 6\lambda + 3 = 0$

$\Rightarrow \boxed{\lambda = \frac{-1}{2}}$

$P\left(\frac{1}{2}, \frac{-1}{2}, 1\right)$

Let another line L' : $\frac{x+1}{1} = \frac{y-1}{-1} = \frac{z+2}{1}$

General point on line L' is $(\mu - 1, -\mu + 1, \mu - 2)$

Point of intersection of line L and L' is

$\lambda + 1 = \mu - 1$	$2\lambda + 2 = \mu - 2$
$\Rightarrow \mu - \lambda = 2 \ldots. (1)$	$\Rightarrow 2\lambda = \mu - 4$

$\Rightarrow \boxed{\lambda = -2}$ and $\boxed{\mu = 0}$

$Q(-1, 1, -2)$

$2(PQ)^2 = 2\left(\left(\frac{1}{2}+1\right)^2 + \left(\frac{-1}{2}-1\right)^2 + (1+2)^2\right)$

$= 2\left(\frac{9}{4} + \frac{9}{4} + 9\right)$

$= 27$

15. If $\sin x + \sin^2 x = 1,\ x \in \left(0, \frac{\pi}{2}\right)$, then

$(\cos^{12} x + \tan^{12} x) + 3(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x)$

$+(\cos^6 x + \tan^6 x)$ is equal to

(1) 2 (2) 1

(3) 3 (4) 4

Answer (1)

Sol. $\sin x + \sin^2 x = 1,\ x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow \sin x = \cos^2 x \Rightarrow \tan x = \cos x$

$\left(\cos^{12} + \tan^{12} x\right) + 3\left(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x\right)$

$+(\cos^6 x + \tan^6 x)$

$= (\cos^{12} x + \cos^{12} x) + 3(\cos^{10} x + \cos^{10} x$

$+\cos^8 x + \cos^8 x) + (\cos^6 x + \cos^6 x)$

$= 2\cos^{12} x + 6(\cos^{10} x + \cos^8 x) + 2\cos^6 x$

$= 2[\cos^{12} x + 3(\cos^{10} x + \cos^8 x) + \cos^6 x]$

$= 2[(\sin^2 x)^3 + 3\sin^4 x\cos^2 x + 3\sin^2 x\cos^4 x$

$+(\cos^2 x)^3]$

$= 2[(\sin^2 x + \cos^2 x)^3] = 2$



16. Let $\alpha, \beta\ (\alpha \neq \beta)$ be the values of m, for which the equations $x + y + z = 1$; $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10}\left(n^{\alpha} + n^{\beta}\right)$ is equal to

(1) 560 (2) 440

(3) 3080 (4) 4410

Answer (2)

Sol. $\Delta = 0, \Delta_x = 0, \Delta_y = 0, \Delta_z = 0$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & m \\ 1 & 4 & m^2 \end{vmatrix} = 0$$

$1(2m^2 - 4m) - 1(m^2 - m) + 1(4 - 2) = 0$

$m^2 - 3m + 2 = 0$

$(m - 1)(m - 2) = 0$

$m = 1, 2$

$$\Delta_x = \begin{vmatrix} 1 & 1 & 1 \\ m & 2 & 4 \\ m^2 & 4 & 10 \end{vmatrix} = 0 \Rightarrow m = 1, 2$$

$$\Delta_y = \begin{vmatrix} 1 & 1 & 1 \\ 1 & m & 4 \\ 1 & m^2 & 10 \end{vmatrix} = 0 \Rightarrow m = 1, 2$$

$$\sum_{x=1}^{10}(n^{\alpha} + n^{\beta}) = \sum_{n=1}^{10}(n^1 + n^2) = \frac{n(n+1)}{2} + \frac{n(n+1)(2n+1)}{6}$$

$= 55 + 385 = 440$

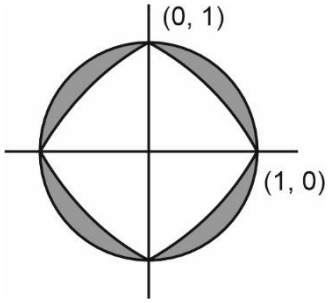
17. Let the area enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ be α. If $9\alpha = \beta\pi + \gamma; \beta, \gamma$ are integers, then the value of $|\beta - \gamma|$ equals.

(1) 18 (2) 33

(3) 27 (4) 15

Answer (2)

Sol.



Required area $= \pi - 4\int_0^1 (1 - x^2)dx$

$$= \pi - 4\left[x - \frac{x^3}{3}\right]_0^1$$

$$= \pi - 4 \times \frac{2}{3} = \pi - \frac{8}{3}$$

$$\therefore\ \alpha = \pi - \frac{8}{3}$$

$9\alpha = 9\pi - 24 \rightarrow \beta = 9, \gamma = -24$

$|\beta - \gamma| = |9 + 24| = 33$

18. Let $\hat{a}$ be a unit vector perpendicular to the vectors $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} - \hat{k}$, and makes an angle of $\cos^{-1}\left(-\frac{1}{3}\right)$ with the vector $\hat{i} + \hat{j} + \hat{k}$. If a makes an angle of $\frac{\pi}{3}$ with the vector $\hat{i} + \alpha\hat{j} + \hat{k}$, then the value of α is

(1) $\sqrt{6}$ (2) $-\sqrt{3}$

(3) $-\sqrt{6}$ (4) $\sqrt{3}$

Answer (3)

Sol. $$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 3 & -1 \end{vmatrix} = \hat{i}(-7) + 7\hat{j} + 7\hat{k}$$

$$\hat{a} = \pm\frac{\left(-7\hat{i} + 7\hat{j} + 7\hat{k}\right)}{\sqrt{7^2 + 7^2 + 7^2}} = \pm\left(\frac{-\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}\right)$$

Now, $\cos\theta = \pm\frac{(-1+1+1)}{\sqrt{3}\cdot\sqrt{3}} = \pm\frac{1}{3}$

$$\Rightarrow \cos^{-1}\left(\frac{-1}{3}\right) \Rightarrow \hat{a} = \frac{-(-\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$$

$$\hat{a} = \frac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$$



$$\cos\frac{\pi}{3}=\frac{1-\alpha-1}{\sqrt{3}.\sqrt{\alpha^2+2}}$$

$$\frac{1}{2}=\frac{-\alpha}{\sqrt{3}.\sqrt{\alpha^2+2}}\rightarrow \alpha<0$$

$3(\alpha^2 + 2) = 4\alpha^2$

$6 = \alpha^2$

$\alpha = \pm\sqrt{6}$

Clearly, $\alpha = -\sqrt{6}$

19. The remainder, when 7^{103} is divided by 23, is equal to

(1) 9

(2) 6

(3) 17

(4) 14

Answer (4)

Sol. $7, 7^2 = 49, 7^3 = 343 \equiv (-2) \pmod{23}$

$\Rightarrow 7^{102} \equiv (7^3)^{34} \equiv (-2)^{34} \equiv 4^{17} \pmod{23}$

$\Rightarrow 4^6 \equiv 2 \pmod{23}$

$4^{17} \equiv (2)(2)(12) \equiv 2 \pmod{23}$

$7^{103} \equiv 7.4^{17} \equiv 14 \pmod{23}$

Alter : $7^{\phi(23)} \equiv 1 \pmod{23}$, gcd(7, 23) = 1

$\phi(23) = (23 - 1) = 22$

$\Rightarrow 7^{22} \equiv 1 \pmod{23} \Rightarrow 7^{11} \equiv (-1) \bmod(23)$

[as $7^{11} \not\equiv 1 \bmod 23$] $\Rightarrow 7^{99} \equiv -1 \bmod(23)$

$7^{102} \equiv 2 \pmod{23}$

$\Rightarrow 7^{103} \equiv 14 \bmod 23$

20. If the set of all $a \in \mathbf{R}$, for which the equation $2x^2 + (a-5)x + 15 = 3a$ has no real root, is the interval (α, β) and $X = \{x \in Z : \alpha < x < \beta\}$, then $\sum_{x \in X} x^2$ is equal to

(1) 2129

(2) 2119

(3) 2139

(4) 2109

Answer (3)

Sol. $(a-5)^2 - 8(15 - 3a) < 0$

$\Rightarrow (a + 19)(a - 5) < 0$

$a \in (-19, 5)$

Hence, $x \in (-19, 5)$

$\sum x_i^2 = (1^2 + 2^2 + \ldots 4^2) + (1^2 + 2^2 + \ldots 18^2)$

= 2139

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. If $24\int_0^{\frac{\pi}{4}}\left(\sin\left|4x-\frac{\pi}{12}\right|+[2\sin x]\right)dx = 2\pi + \alpha$, where $[\cdot]$ denotes the greatest integer function, then α is equal to_____.

Answer (12)

Sol. Let $I = 24\int_0^{\frac{\pi}{2}}\left(\sin\left|4x-\frac{\pi}{2}\right|+[2\sin x]\right)dx$ ...(i)

Now $\left|4x-\frac{\pi}{12}\right| = \begin{cases} -4x+\frac{\pi}{12} & ; \ x < \frac{\pi}{48} \\ 4x-\frac{\pi}{12} & ; \ x \geq \frac{\pi}{48} \end{cases}$



$\therefore$ from (i)

$$I = 24\int_0^{\frac{\pi}{48}} -\sin\left(4x-\frac{\pi}{12}\right)dx + \int_{\frac{\pi}{48}}^{\frac{\pi}{4}} \sin\left(4x-\frac{\pi}{12}\right)$$

$$+\int_0^{\frac{\pi}{6}} [2\sin x]dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} [2\sin x]dx$$

$$I = 24\left[\frac{\left(1-\cos\frac{\pi}{12}\right)}{4} - \frac{\left(-\cos\frac{\pi}{12}-1\right)}{4}\right] + \frac{\pi}{4} - \frac{\pi}{6}$$

$$I = 24\left(\frac{1}{2}\right) + \frac{\pi}{4} - \frac{\pi}{6}$$

$I = 2\pi + 12 = 2\pi + \alpha$ (from above)

$\therefore \quad \alpha = 12$

22. Let $a_1, a_2, ..., a_{2024}$ be an arithmetic Progression such that

$a_1 + (a_5 + a_{10} + a_{15} + ... + a_{2020}) + a_{2024} = 2233.$

Then $a_1 + a_2 + a_3 + ... + a_{2024}$ is equal to

Answer (11132)

Sol. As $a_1 + a_5 + a_{10} + \ldots + a_{2020} + a_{2024} = 2233$ ...(1)

We know in arithmetic progression.

Sum of terms equidistant from ends is equal

$\therefore$ from (1)

$$\underbrace{a_1 + a_{2024} = a_5 + a_{2020} = a_{10} + a_{2015} = \ldots}_{203 \text{ pairs}}$$

$\Rightarrow$ 203 $(a_1 + a_{2024}) = 2233$

$\Rightarrow$ $a_1 + a_{2024} = 11$

Now $\sum_{i=1}^{2024} a_i = S_{2024} = \frac{2024}{2}[a_1 + a_{2024}]$

$= 1012\ (11)$

$= 11132$

23. Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b), for which $\left|\frac{z-a}{z+b}\right| = 1$ and

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1,\ z \in C,$$ where ω and ω^2 are the roots of $x^2 + x + 1 = 0$, is equal to____.

Answer (10)

Sol. $a, b \in I, -3 \le a, b \le 3; a + b \neq 0$

$|z - a| = |z + b|$

$$\begin{vmatrix} z+1 & w & w^2 \\ w & z+w^2 & 1 \\ w^2 & 1 & z+w \end{vmatrix} = 1$$

$$\Rightarrow \begin{vmatrix} z & z & z \\ w & z+w^2 & 1 \\ w^2 & 1 & z+w \end{vmatrix} = 1$$

$$\Rightarrow z\begin{vmatrix} 1 & 1 & 1 \\ w & z+w^2 & 1 \\ w^2 & 1 & z+w \end{vmatrix} = 1$$

$$\Rightarrow z\begin{vmatrix} 1 & 0 & 0 \\ w & z+w^2-w & 1-w \\ w^2 & 1-w^2 & z+w-w^2 \end{vmatrix} = 1$$

$\Rightarrow z^3 = 1$

$\Rightarrow z = w, w^2, 1$

Now, $|1 - a| = |1 + b|$

$\Rightarrow$ 10 pairs



24. If $\lim\limits_{t\to 0}\left(\int\limits_0^1 (3x+5)^t\,dx\right)^{1/t} = \frac{\alpha}{5e}\left(\frac{8}{5}\right)^{2/3}$, then α is equal to_____.

Answer (64)

Sol. $\frac{\alpha}{5e} = \exp\left(\lim\limits_{t\to 0}\frac{1}{t}\left(\int\limits_0^1 (3x+5)^t\,dx - 1\right)\right)$

$= \exp\left(\lim\limits_{t\to 0}\frac{1}{t}\left(\frac{(3x+5)^{t+1}}{3(t+1)}\right)_0^1 - 1\right)$

$= \exp\left(\lim\limits_{t\to 0}\frac{1}{t}\left(\frac{8^{t+1}-5^{t+1}}{3(t+1)} - 1\right)\right)$

$= \exp\left(\lim\limits_{t\to 0}\frac{1}{t}\left(\frac{8^{t+1}-5^{t+1}-3t-3}{3(t+1)}\right)\right)$

$= \exp\left(\lim\limits_{t\to 0}\left(\frac{8^{t+1}\cdot\ln 8 - 5^{t+1}\ln 5 - 3}{3(t+1)}\right)\right)$

$= \exp\left(\frac{\ln 8^8 - \ln 5^5 - 3}{5}\right)$

$= \left(\frac{8}{5}\right)^{2/3}\frac{\alpha}{5e} = \exp\left(\frac{\ln\left(\frac{8^8}{5^5}\right)}{5} - 1\right)$

$\Rightarrow \left(\frac{8}{5}\right)^{2/3}\frac{\alpha}{5} = \left(\frac{8^8}{5^5}\right)^{1/3} = \left(\frac{8^6\cdot 8^2}{5^3\cdot 5^2}\right)^{1/3} = \frac{64}{5}\left(\frac{8}{5}\right)^{2/3}$

$\Rightarrow \alpha = 64$

25. Let $y^2 = 12x$ be the parabola and S be its focus. Let PQ be a focal chord of the parabola such that $(SP)(SQ) = \frac{147}{4}$. Let C be the circle described taking PQ as a diameter. If the equation of a circle C is $64x^2 + 64y^2 - \alpha x - 64\sqrt{3}y = \beta$, then $\beta - \alpha$ is equal to_____.

Answer (1328)

Sol. $y^2 = 12x$, $a = 3$, $(SP)\,(SQ) = \frac{147}{4}$

Let $P(3t^2, 6t)$ and $t_1t_2 = -1$ (ends of focal chord)

$\because Q \equiv \left(\frac{3}{t^2}, \frac{6}{t}\right)$

$(SP)\,(SQ) = (PM_1)\,(QM_2)$

$= \left(3+3t^2\right)\left(3+\frac{3}{t^2}\right) = \frac{147}{4}$

$\Rightarrow \frac{\left(1+t^2\right)^2}{t^2} = \frac{49}{12}$

$\Rightarrow t^2 = \frac{3}{4}, \frac{4}{3}$

$\Rightarrow t = \pm\frac{\sqrt{3}}{2}, \frac{\pm 2}{\sqrt{3}}$

Considering $t = \frac{-\sqrt{3}}{2}$

$\therefore P\left(\frac{9}{4}, -3\sqrt{3}\right)$ and $Q\left(4, 4\sqrt{3}\right)$

$\therefore$ Equation of circle

$\Rightarrow (x-4)\left(x-\frac{9}{4}\right) + \left(y+3\sqrt{3}\right)\left(y-4\sqrt{3}\right) = 0$

$\Rightarrow x^2 + y^2 - \frac{25}{4}x - \sqrt{3}y - 27 = 0$

$\Rightarrow \alpha = 400, \beta = 1728$

$\therefore \beta - \alpha = 1328$



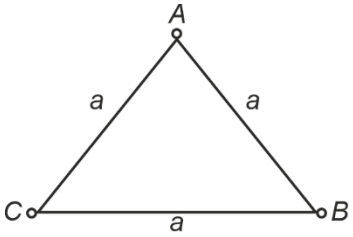
PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26.



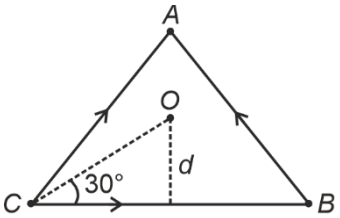
Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in free space. At $t = 0$, they are given an initial velocity $\vec{V_A} = V_0\,\vec{AC}$, $\vec{V_B} = V_0\,\vec{BA}$ and $\vec{V_C} = V_0\,\vec{CB}$. Here, $\vec{AC}$, $\vec{CB}$ and $\vec{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is :

(1) $\frac{1}{2}amV_0$ (2) $\frac{3}{2}amV_0$

(3) $\frac{\sqrt{3}}{2}amV_0$ (4) $3amV_0$

Answer (3)

Sol.



$d = \frac{a}{2\sqrt{3}}$

Angular momentum of one mass about point O

$L = mvd$

$= mv_0 \cdot \frac{a}{2\sqrt{3}}$

Net angular momentum about point O

$L_{net} = 32$

$= \frac{\sqrt{3}mv_0a}{2}$

27. Two bodies A and B of equal mass are suspended from two massless springs of spring constant k_1 and k_2, respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is

(1) $\sqrt{\frac{k_1}{k_2}}$ (2) $\frac{k_1}{k_2}$

(3) $\frac{k_2}{k_1}$ (4) $\sqrt{\frac{k_2}{k_1}}$

Answer (1)

Sol. Here $\omega = \sqrt{\frac{k}{m}}$

and maximum velocity $V = A\omega = A\sqrt{\frac{k}{m}}$

So, $\frac{V_A}{V_B} = \sqrt{\frac{k_1}{k_2}}$

28. A point charge causes an electric flux of -2×10^4 Nm^2C^{-1} to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is :

(Given $\in_0 = 8.85 \times 10^{-12}$ $C^2N^{-1}m^{-2}$)

(1) 17.7×10^{-8} C (2) -15.7×10^{-8} C

(3) 15.7×10^{-8} C (4) -17.7×10^{-8} C

Answer (4)

